

Project Report: Healthcare Appointment No-Show Prediction

Executive Summary:

Missing medical appointments ("no-shows") places a heavy strain on healthcare systems by wasting clinical resources, increasing wait times, and delaying patient care. This project implements a data-driven approach using Python predictive modeling and Power BI dashboards to anticipate missed appointments, isolate behavioral drivers, and provide actionable scheduling optimizations.

Objectives & Data Strategy:

- **Predictive Identification:** Proactively flag high-risk appointments before they occur.
- **Driver Analysis:** Evaluate how lead time (days between booking and the appointment), patient demographics, and SMS reminders impact attendance.
- **Operational Cleanup:** Filter out data anomalies, standardize categorical health factors, and engineer time-based features (weekdays, operational shifts) to build a robust, clean dataset.

Predictive Modeling & Dashboards:

- **The Decision Tree Model:** A rule-based machine learning model segments patients based on historical patterns (e.g., if lead time is long and no SMS was sent, risk increases). Its transparent, flow-chart logic allows healthcare staff to easily understand and trust the risk factors behind every prediction.
- **Power BI Business Intelligence:** A dynamic dashboard translates these predictions into visual operational insights. It tracks real-time no-show rates, highlights peak risk hours via temporal heatmaps, and monitors the direct impact of text reminders on attendance trends.

Strategic Recommendations:

- **Tiered Interventions:** Automate standard text reminders for low-risk slots, but flag high-risk patients for direct, personal phone calls from clinic staff forty-eight hours in advance.
- **Calculated Overbooking:** Implement strategic double-booking or integrate flexible walk-in buffers specifically during historical peak no-show

windows (such as mid-week afternoon slots) to maintain high clinical utilization.

- ***Smart Waitlists:*** Connect the prediction engine to an automated waitlist. If a booked slot crosses a high-risk threshold, automatically text waitlisted patients a priority offer to fill the potential gap early.

Deliverables: Power BI, Python